

Java Topics for 2 Years Experience

Core Java (Must-Know)

- OOP Principles (Encapsulation, Inheritance, Polymorphism, Abstraction)
- Data Types & Operators
- Control Statements (if, switch, loops)
- Strings & StringBuilder vs StringBuffer
- Exception Handling
- Access Modifiers (private, protected, etc.)
- Static, Final, This, Super
- Inner Classes

Collections Framework

- List, Set, Map, Queue
- HashMap vs TreeMap vs LinkedHashMap
- ArrayList vs LinkedList
- HashSet vs TreeSet
- Comparator vs Comparable
- Fail-fast vs Fail-safe
- Iterators (Iterator vs ListIterator)

Multithreading & Concurrency

- Thread class vs Runnable
- Thread lifecycle
- Synchronized, Locks
- wait(), notify(), notifyAll()
- Thread Pool (ExecutorService)
- Race conditions and deadlocks

Java 8 Features

- Lambda Expressions
- Streams API (map, filter, reduce, collect, etc.)
- Functional Interfaces (Predicate, Function, etc.)
- Method References
- Optional class

- Default & Static methods in interfaces

Java Memory Management

- Heap vs Stack
- Garbage Collection Basics
- Strong vs Weak references
- Memory leaks in Java

File I/O & Serialization

- BufferedReader, FileReader, Scanner
- Serialization & Deserialization
- Try-with-resources

Unit Testing (Basic)

- JUnit / TestNG
- Assertions
- Mocking (Mockito basics)

Design Patterns (Basic)

- Singleton
- Factory
- Observer
- Strategy

SQL Basics

- Joins, Aggregations, Subqueries
- CRUD operations

Spring Boot / Spring (If applicable)

- @Autowired, @Component, @Service, @Repository
- Dependency Injection
- REST API (GET, POST, PUT, DELETE)
- Controller-Service-Repository structure
- Spring Data JPA basics

Miscellaneous

- equals() vs ==
- hashCode() and equals() contract

- Immutability (how to make a class immutable)
- Wrapper classes, Autoboxing
- Enums
- Java vs JavaScript (if full stack)
- How exceptions affect performance